Sensing and the Imaging Process (V)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

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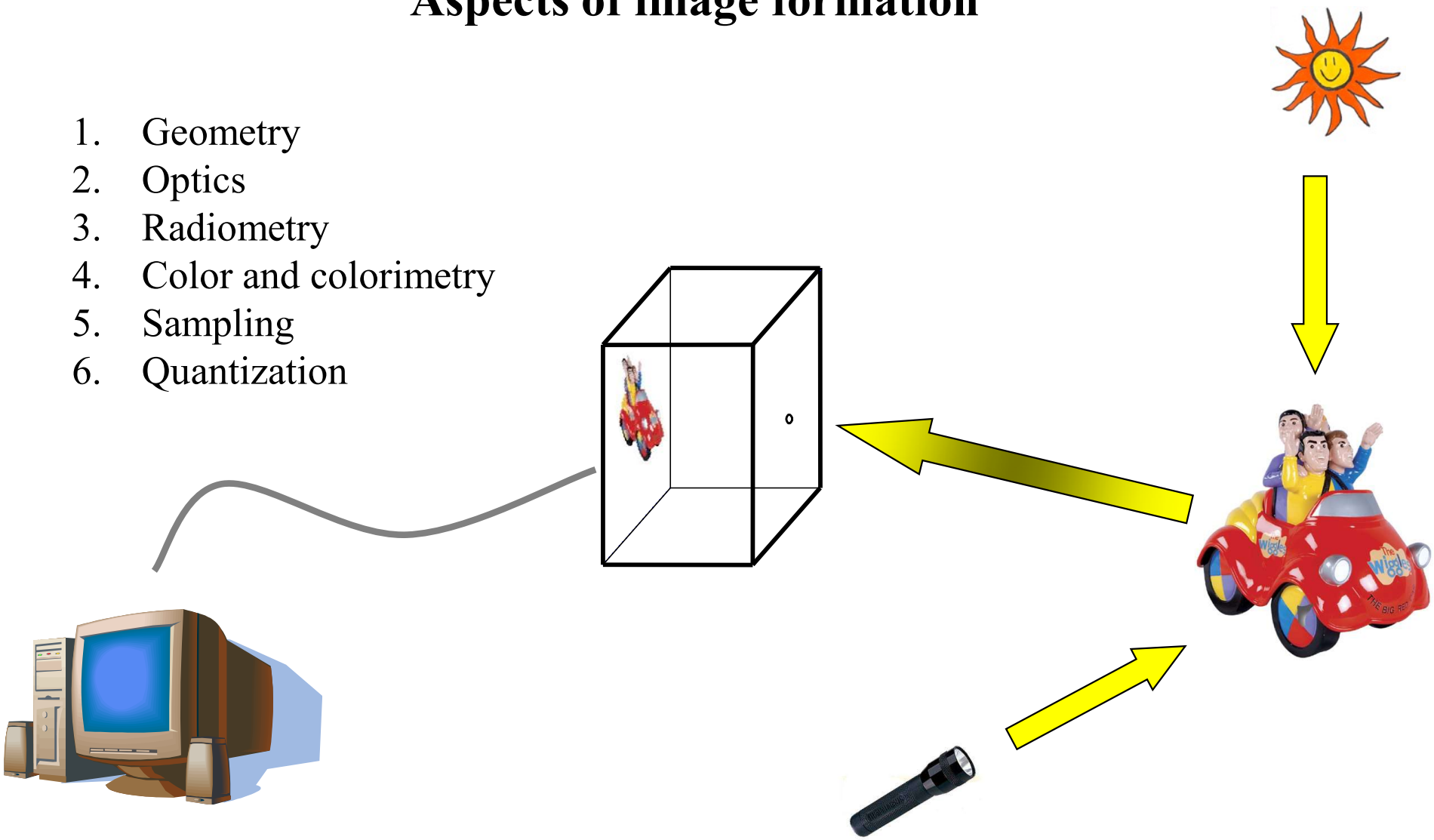
Image formation and representation

Aspects of image formation

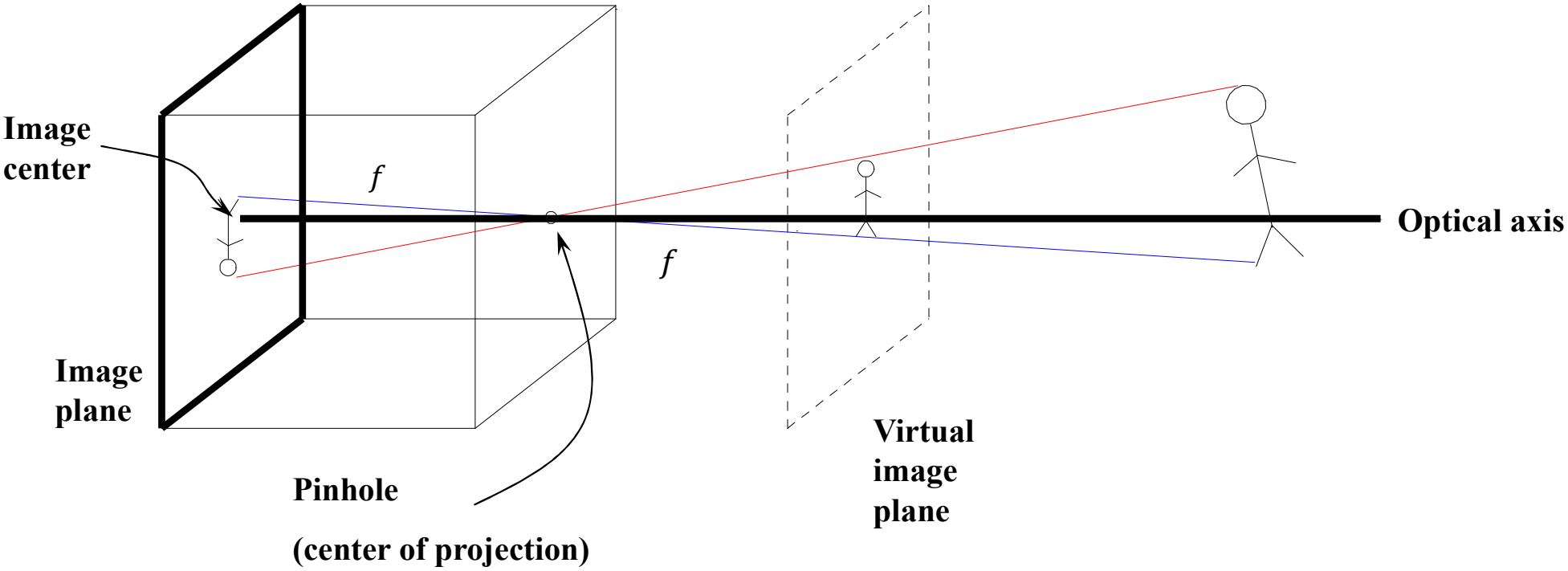
1. Geometry
2. Optics
3. Radiometry
4. Color and colorimetry
5. Sampling
6. Quantization



Geometry of Image Formation

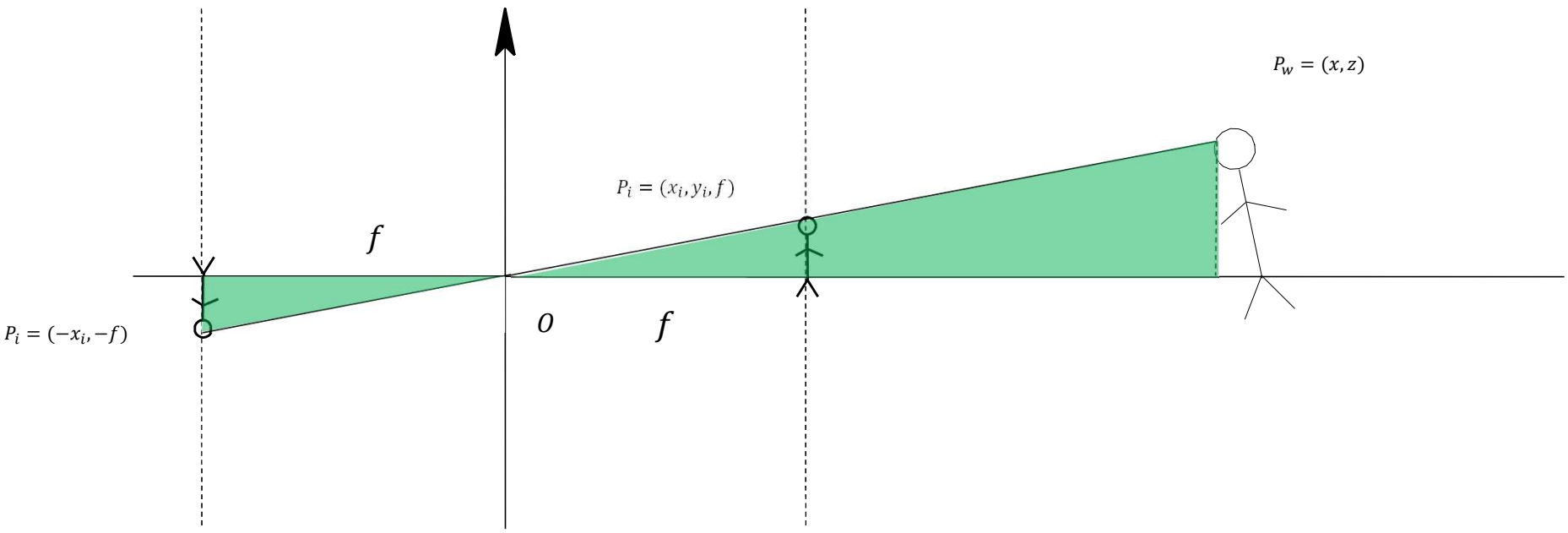
Geometry of Image Formation

The geometry of image formation – The Pinhole camera model



Geometry of Image Formation

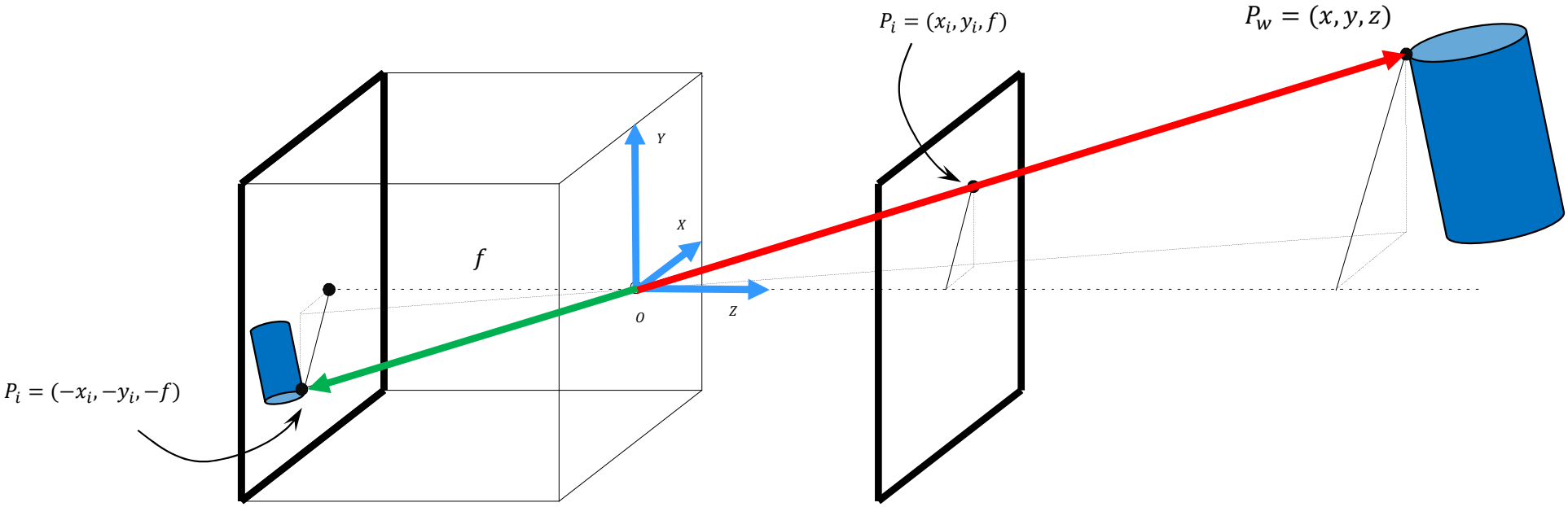
Pinhole camera model: Perspective projection



$$\frac{x_i}{f} = \frac{x}{z}$$

Geometry of Image Formation

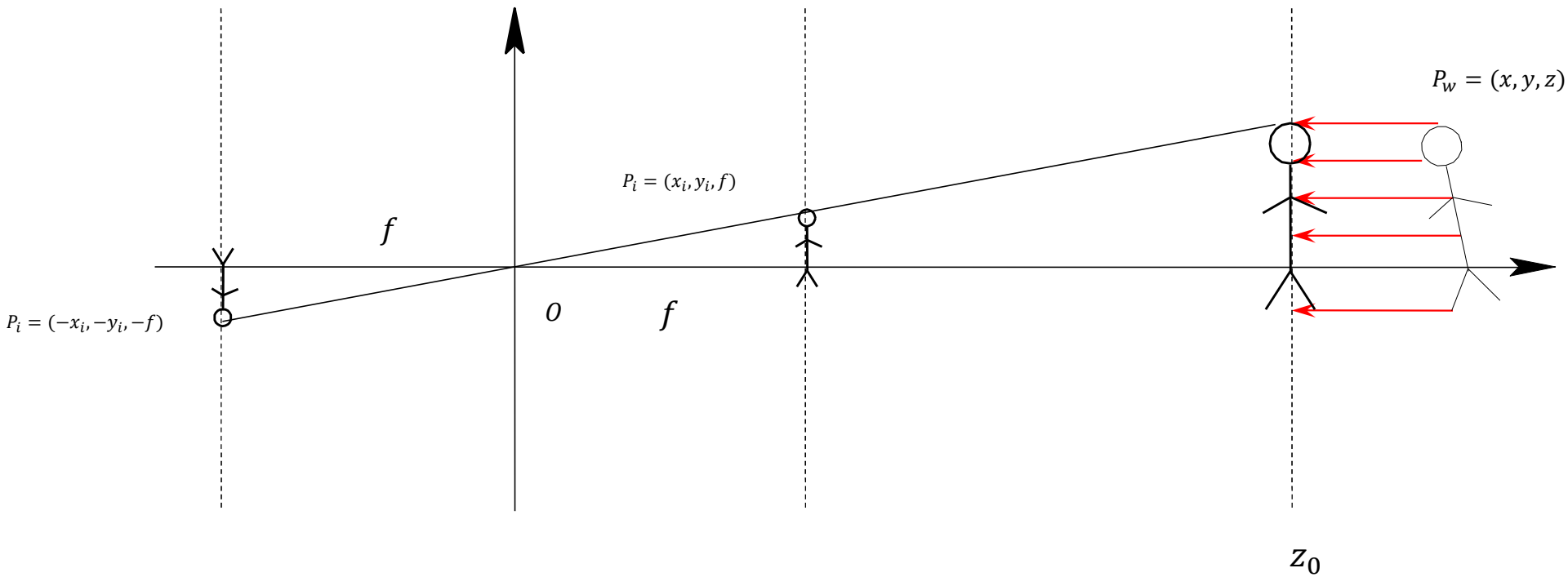
Pinhole camera model: Perspective projection



$$\overrightarrow{OP_i} = \lambda \cdot \overrightarrow{OP_w} \quad \Rightarrow \quad \lambda = \frac{x_i}{x} = \frac{y_i}{y} = \frac{f}{z} \quad \Rightarrow \quad \begin{cases} x_i = f \cdot \dfrac{x}{z} \\ y_i = f \cdot \dfrac{y}{z} \end{cases}$$

Geometry of Image Formation

Linear approximation: Weak perspective projection

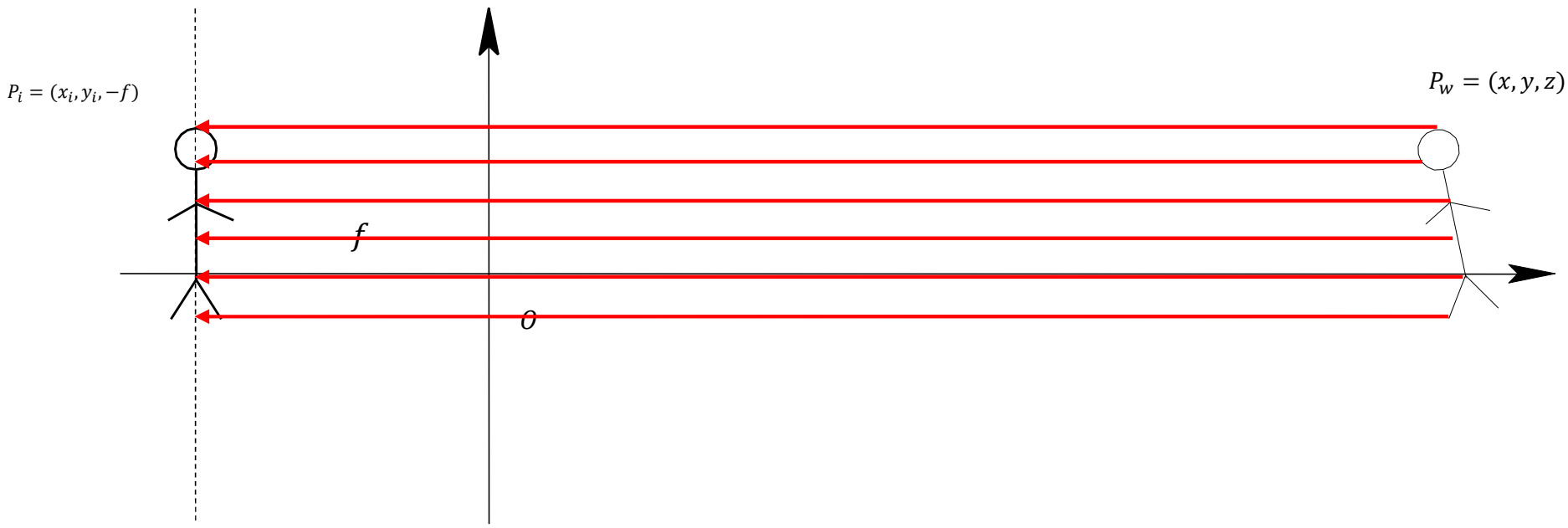


$$x_i = \frac{f}{z} x = s \cdot x$$
$$y_i = \frac{f}{z} y = s \cdot y$$

$$\Leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Geometry of Image Formation

Linear approximation: Orthographic projection



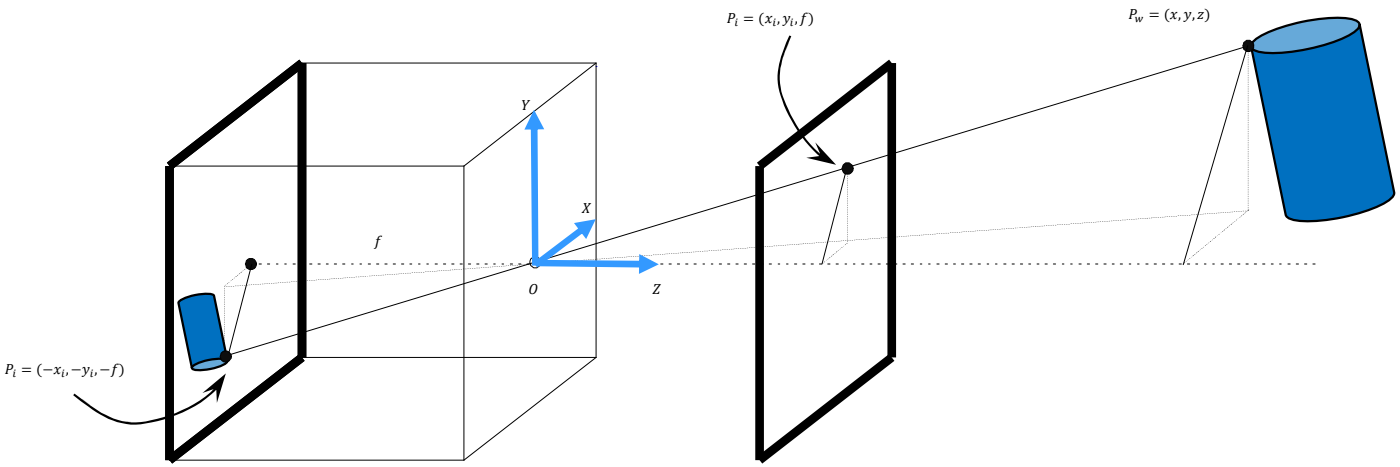
$$\begin{aligned} x_i &= x \\ y_i &= y \end{aligned} \quad \Leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \end{pmatrix} = I \begin{pmatrix} x \\ y \end{pmatrix}$$

Geometry of Image Formation

August Ferdinand Möbius

Möbius strip

Homogeneous coordinates

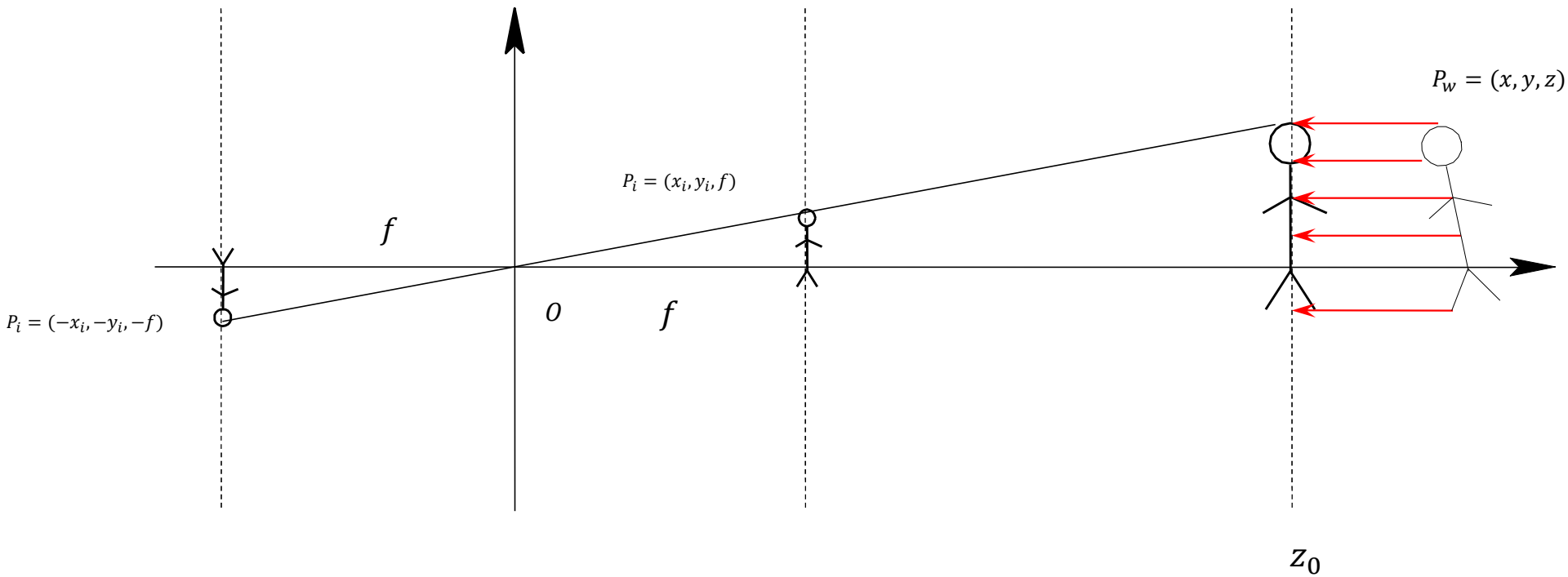


$$(x, y) \quad \Leftrightarrow \quad [x \cdot w, y \cdot w, w] \qquad w \neq 0$$
$$(x, y, z) \quad \Leftrightarrow \quad [x \cdot w, y \cdot w, z \cdot w, w] \qquad w \neq 0$$

most often w **is set as** $w = 1$

Geometry of Image Formation

Weak perspective projection in homogeneous coordinates



$$\left.\begin{aligned} x_i &= s \cdot x \\ y_i &= s \cdot y \end{aligned}\right\} \quad \leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

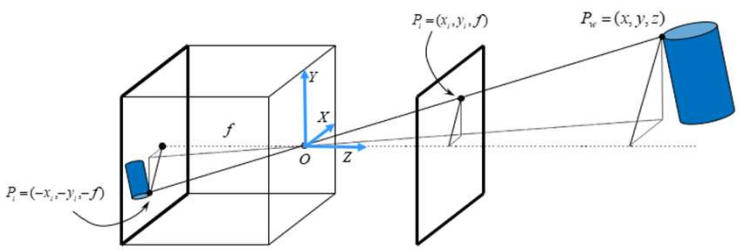
Geometry of Image Formation

Perspective projection in homogeneous coordinates

$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \quad \leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Geometry of Image Formation

Perspective projection in homogeneous coordinates



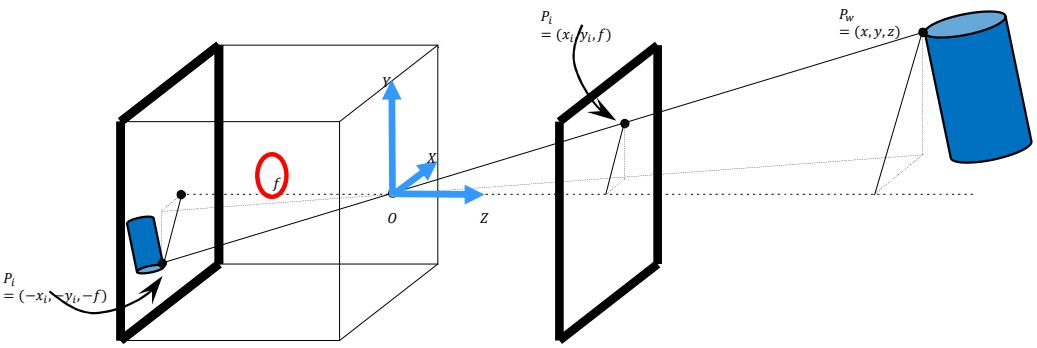
$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \quad \leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \quad \leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \\ f \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ f \cdot z \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\left.\begin{aligned} x_i &= f \cdot \frac{x}{z} \\ y_i &= f \cdot \frac{y}{z} \end{aligned}\right\} \quad \leftrightarrow \quad \begin{pmatrix} x_i \\ y_i \\ f \end{pmatrix} = \begin{bmatrix} x \\ y \\ z \\ z/f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Geometry of Image Formation

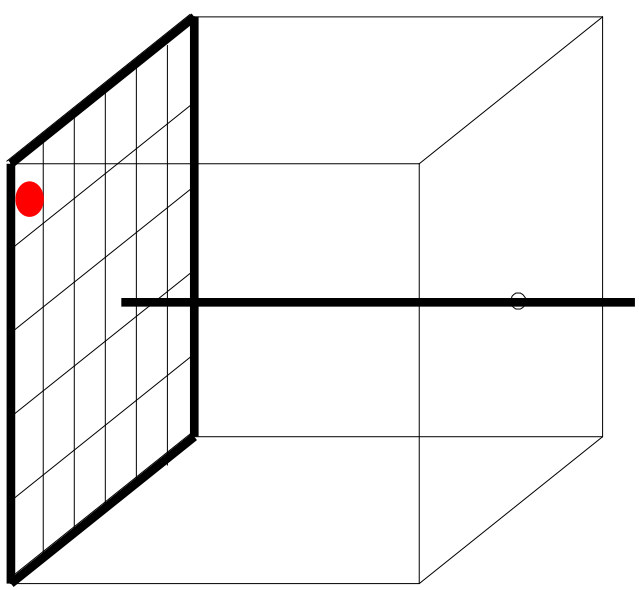
Camera *intrinsic* parameters



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$x_i = S_x f \cdot \frac{x}{z} = \alpha \frac{x}{z} + x_0$$

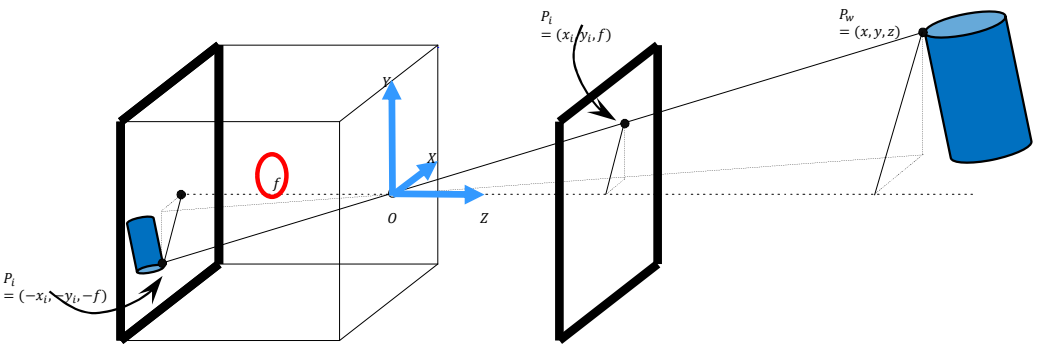
$$y_i = S_y f \cdot \frac{y}{z} = \beta \frac{y}{z} + y_0$$



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} \alpha \cdot x + x_0 z \\ \beta \cdot y + y_0 z \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{The } \textit{calibration matrix } \mathbf{K}} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Geometry of Image Formation

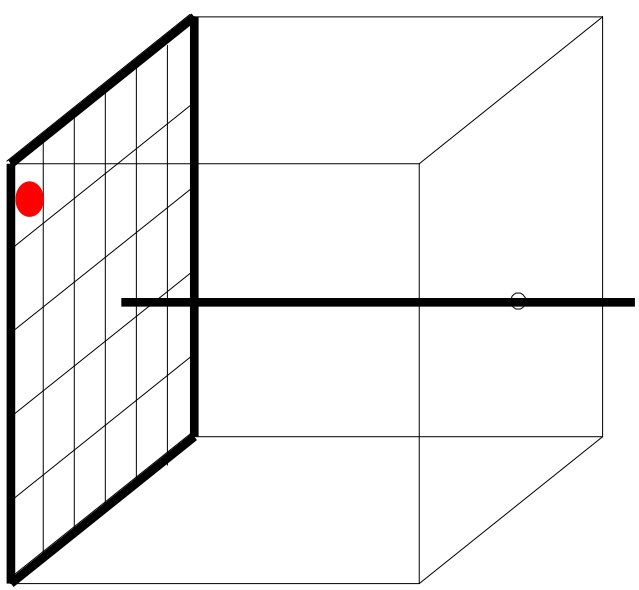
Camera *intrinsic* parameters



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$x_i = S_x f \cdot \frac{x}{z} \quad = \alpha \frac{x}{z} + x_0$$

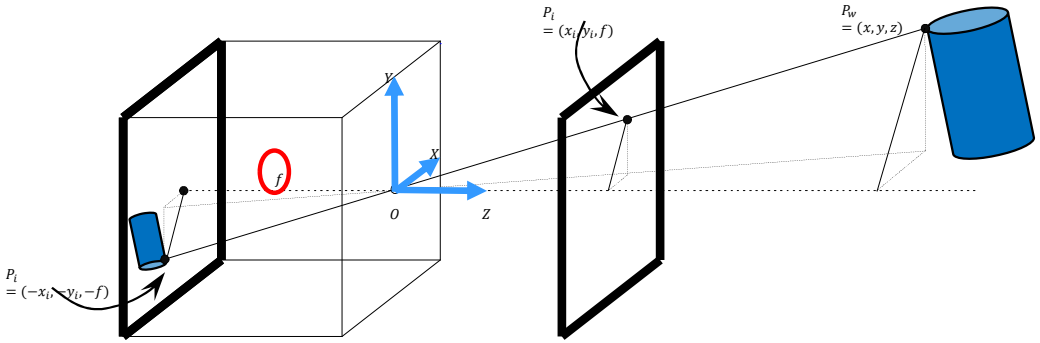
$$y_i = S_y f \cdot \frac{y}{z} \quad = \beta \frac{y}{z} + y_0$$



$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} f \cdot x \\ f \cdot y \\ z \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{The } \textit{calibration matrix } \mathbf{K}} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Geometry of Image Formation

Camera calibration = Recovering the parameters of an unknown camera



$$P_i{}^j = \begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = K \cdot P_w{}^j$$

- Measure enough pairs $\left(P_i{}^j; P_w{}^j\right) \quad j = 1..N$
- Obtain enough linear equations in the parameters of K
- Solve for the parameters of K

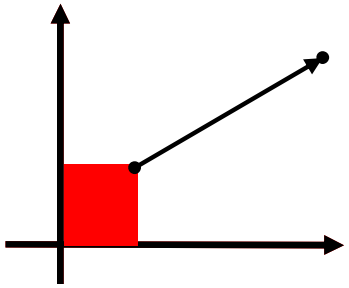
Problem: measuring $P_i{}^j$ is easy, but how can we measure $P_w{}^j$?

Geometry of Image Formation

2D Transformations

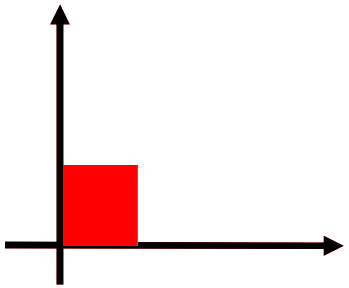
Translation $P_t = P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



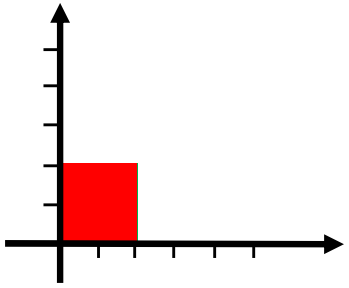
Rotation $P_t = R \cdot P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Scaling $P_t = s\, P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = s \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

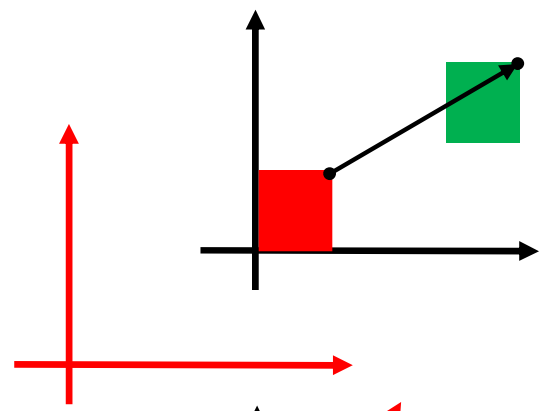


Geometry of Image Formation

2D Transformations

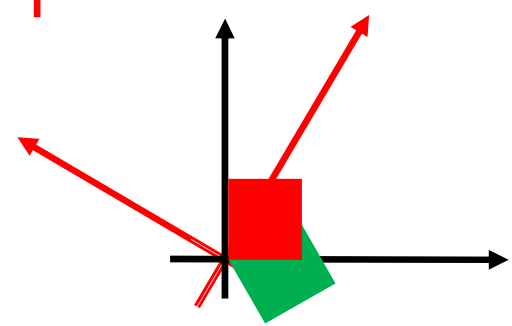
Translation $P_t = P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



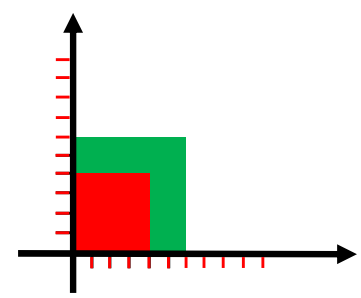
Rotation $P_t = R \cdot P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Scaling $P_t = s\, P_s$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = s \begin{pmatrix} x_s \\ y_s \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

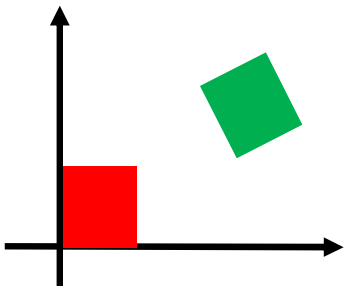


Geometry of Image Formation

2D Transformations

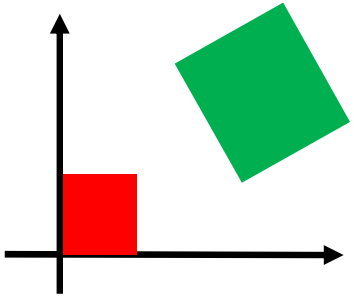
Euclidean $P_t = R \cdot P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Similarity $P_t = s\,R \cdot P_s + \vec{t}$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = s \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \qquad \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} sR & \vec{t} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Geometry of Image Formation

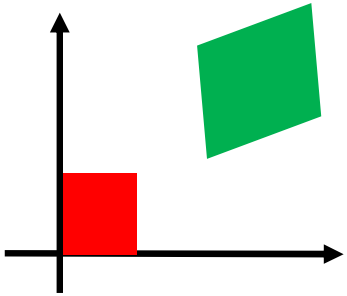
2D Transformations

Affine

$$P_t = A \cdot P_s + \vec{t}$$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \begin{pmatrix} x_s \\ y_s \end{pmatrix} + \begin{pmatrix} a_{02} \\ a_{12} \end{pmatrix}$$

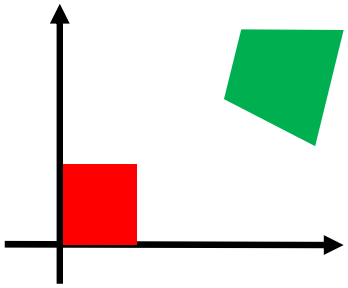
$$\begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



Projective
or
Homography

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \dfrac{h_{00}x_s + h_{01}y_s + h_{02}}{h_{20}x_s + h_{21}y_s + h_{22}} \\ \\ \dfrac{h_{00}x_s + h_{01}y_s + h_{02}}{h_{20}x_s + h_{21}y_s + h_{22}} \end{pmatrix}$$

$$\begin{bmatrix} x_t \\ y_t \\ w \end{bmatrix} = \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$



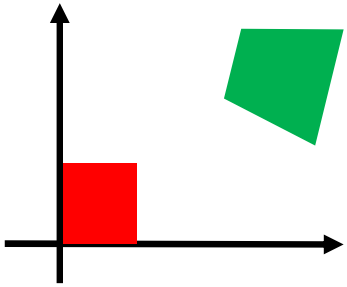
Geometry of Image Formation

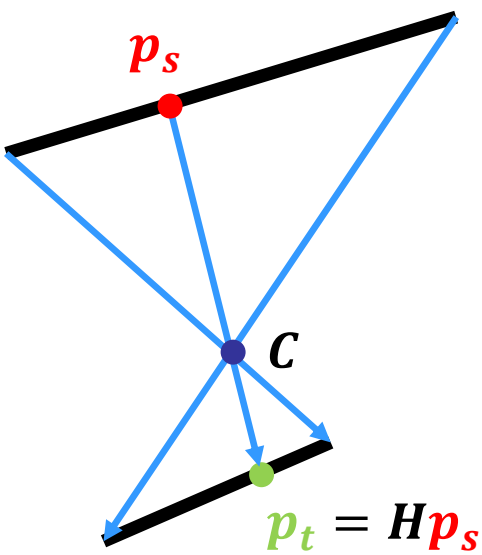
2D Transformations

Projective
or
Homography

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \dfrac{h_{00}x_s + h_{01}y_s + h_{02}}{h_{20}x_s + h_{21}y_s + h_{22}} \\ \\ \dfrac{h_{00}x_s + h_{01}y_s + h_{02}}{h_{20}x_s + h_{21}y_s + h_{22}} \end{pmatrix}$$

$$\begin{bmatrix} x_t \\ y_t \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix}}_{H} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$





Properties:

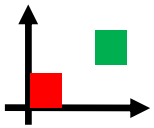
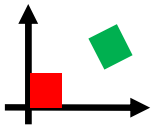
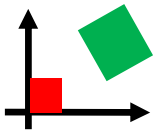
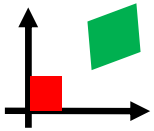
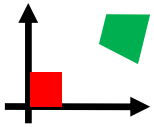
- Maps straight lines to straight lines
- Origin does not necessarily map the origin
- Parallel lines do not necessarily remain parallel
- Closed under composition
- Unique up top a constant
- "Scale" can be fixed by norm constraint, say setting

$$\|H\|_F = \sqrt{\sum\sum (h_{ij})^2} = 1$$

- Hence, in practice, only 8 DoF

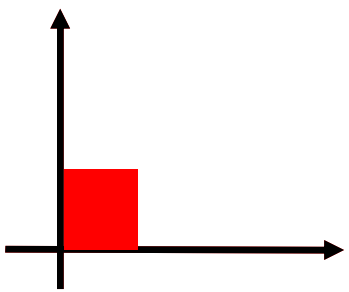
Geometry of Image Formation

2D Transformations

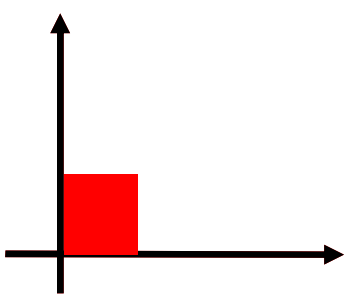
Transformation	Matrix	#DoF	
Translation	$T^T = \begin{bmatrix} I_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	2	
Euclidean	$T^E = \begin{bmatrix} R_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	3	
Similarity	$T^S = \begin{bmatrix} sR_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	4	
Affine	$T^A = \begin{bmatrix} A_{2\times2} & \vec{t}_{2x1} \\ 0 & 1 \end{bmatrix}$	6	
Projective	$T^P = H = [H_{3\times3}]$	8	

Geometry of Image Formation

Sequence of 2D Transformations



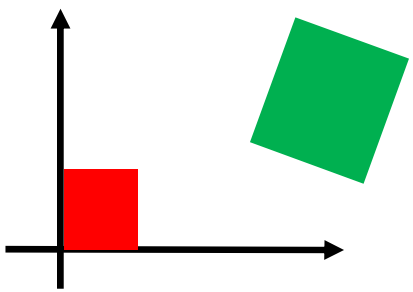
p_s



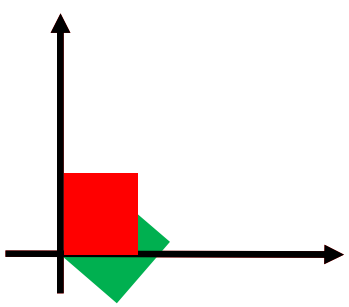
p_s

Geometry of Image Formation

Sequence of 2D Transformations



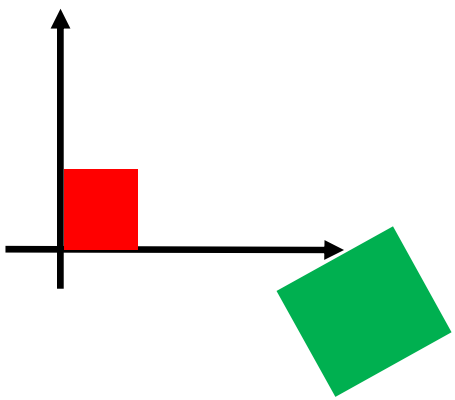
$$p_t = T^1 p_s$$



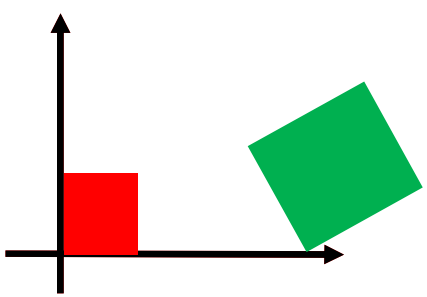
$$p_t = T^2 p_s$$

Geometry of Image Formation

Sequence of 2D Transformations



$$p_t = T^2 T^1 p_s$$



$$p_t = T^1 T^2 p_s$$

Thanks to linearity in homogeneous coordinates:

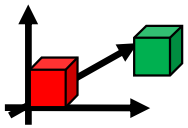
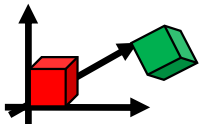
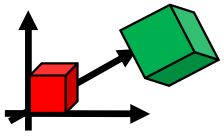
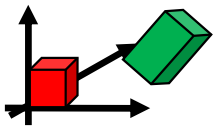
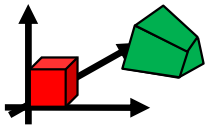
- Transformations can be cascaded

$$p_t = T^n \cdots T^2 T^1 p_s$$

- Complex transformations may be represented as a cascade of simpler ones.
- Order (right to left) of application matters!!

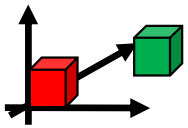
Geometry of Image Formation

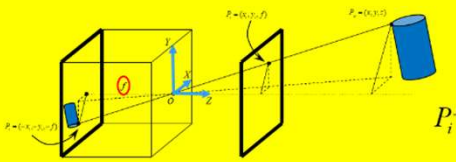
3D Transformations

Transformation	Matrix	#DoF	
Translation	$T^T = \begin{bmatrix} I_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	3	
Euclidean	$T^E = \begin{bmatrix} R_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	6	
Similarity	$T^S = \begin{bmatrix} sR_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	7	
Affine	$T^A = \begin{bmatrix} A_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	12	
Projective	$T^P = H = [H_{4\times4}]$	15	

Geometry of Image Formation

3D Transformations

Transformation	Matrix	#DoF	
Translation	$T^T = \begin{bmatrix} I_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	3	
Euclidean	$T^E = \begin{bmatrix} R_{3\times3} & \vec{t}_{3x1} \\ 0 & 1 \end{bmatrix}$	6	
Similarity	$T^S = \begin{bmatrix} sR_{3\times} \end{bmatrix}$		
Affine	$T^A = \begin{bmatrix} A \end{bmatrix}$		
Projective	$T^P = H = [H$		



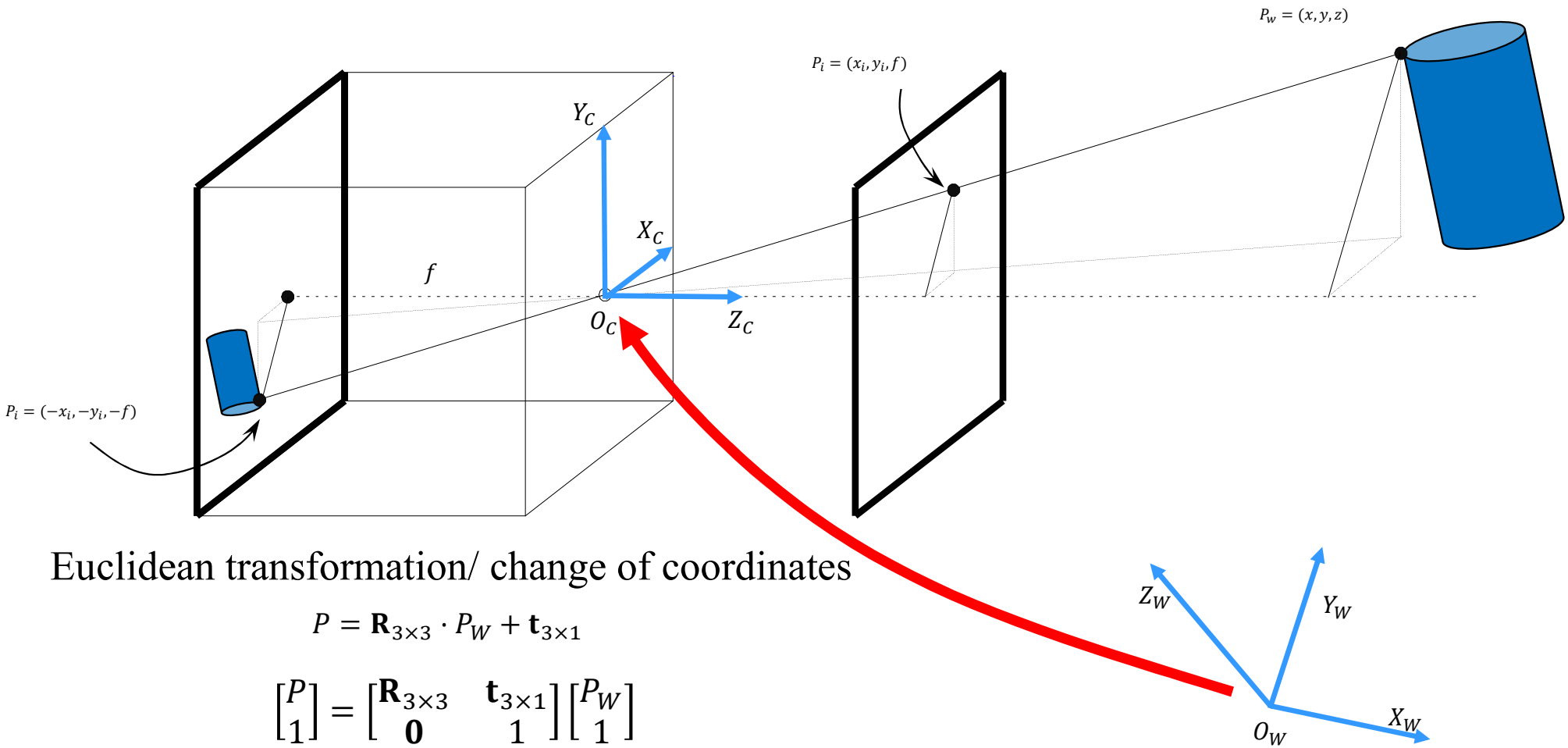
$$P_i^j = \begin{bmatrix} \alpha & 0 & x_0 & 0 \\ 0 & \beta & y_0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = K \cdot P_w^j$$

- Measure enough pairs $\left(P_i^j ; P_w^j\right)$ $j = 1..N$
- Obtain enough linear equations in the parameters of K
- Solve for the parameters of K

Problem : measuring P_i^j is easy, but how can we measure P_w^j?

Geometry of Image Formation

Camera *extrinsic* parameters



Euclidean transformation/ change of coordinates

$$P = \mathbf{R}_{3\times3} \cdot P_W + \mathbf{t}_{3\times1}$$

$$\begin{bmatrix} P \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{R}_{3\times3} & \mathbf{t}_{3\times1} \\ \mathbf{0} & 1 \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} P_W \\ 1 \end{bmatrix}$$

Geometry of Image Formation

The perspective camera projection matrix

World coordinates in camera coordinate frame

$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \boldsymbol{K} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

World coordinates in a world coordinate frame

$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \boldsymbol{KF} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \underbrace{\boldsymbol{M}}_{} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

The ***perspective camera projection matrix*** *M*

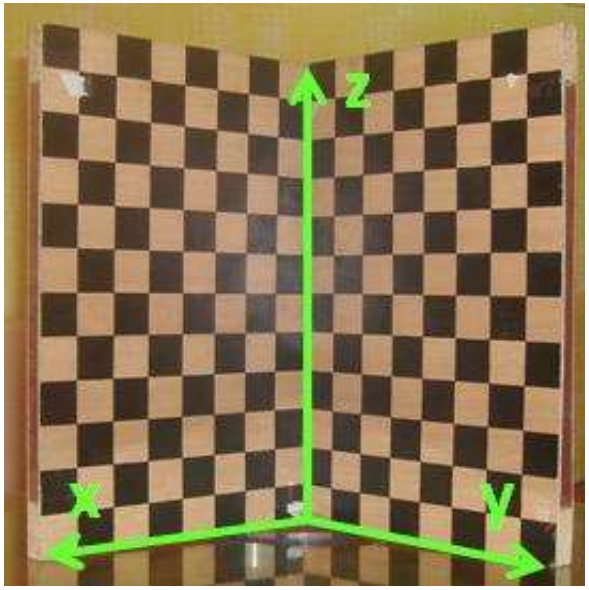
Geometry of Image Formation

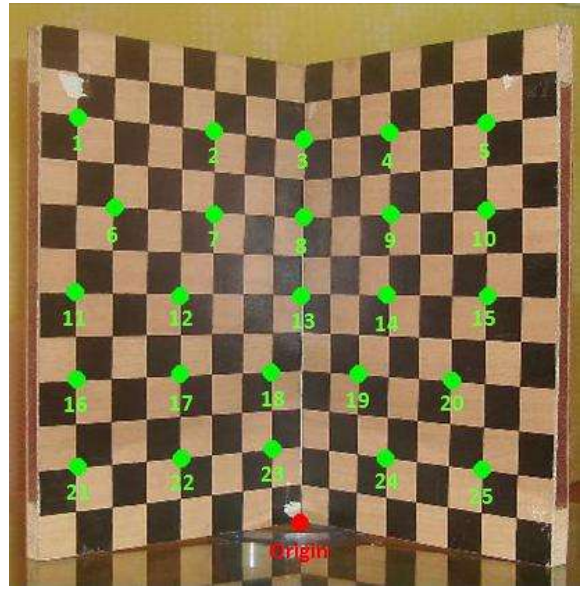
Camera calibration

- Measure enough pairs $\left(P_i{}^j; P_w{}^j\right) \quad j = 1..N \geq 6$

$$P_i{}^j = \mathbf{M} \cdot P_w{}^j$$

- Estimate **M**
- If needed, decompose M=RQ to estimate the intrinsic and extrinsic parameters separately

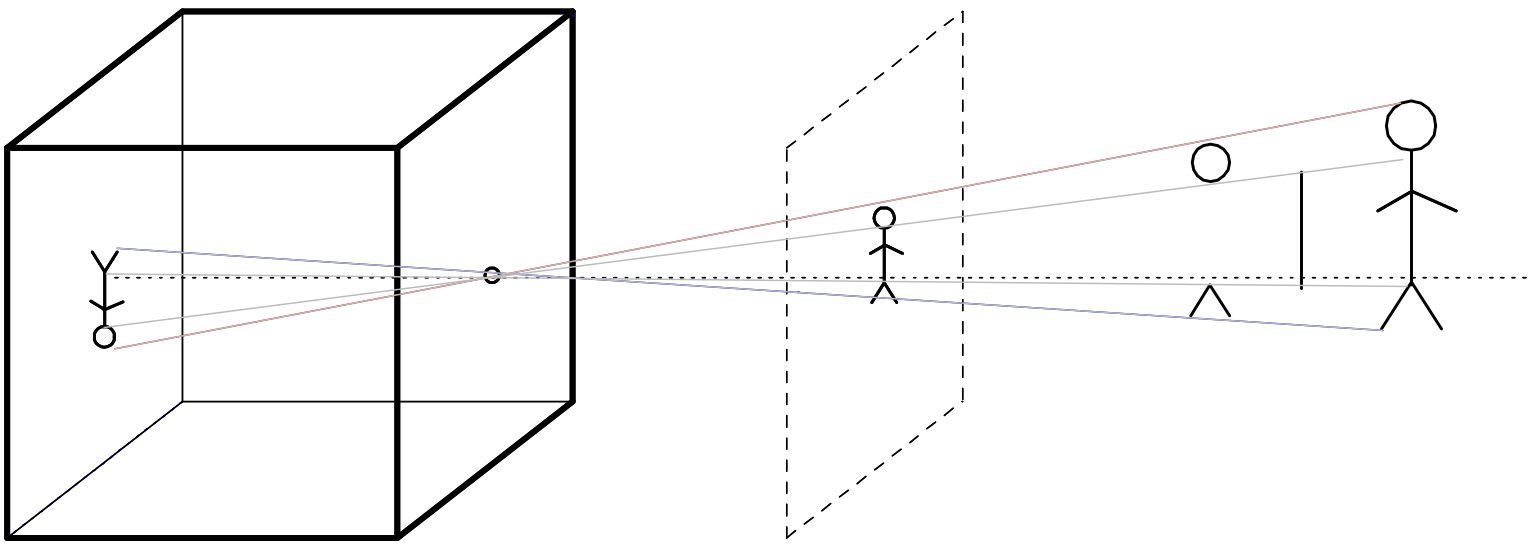




Geometry of Image Formation

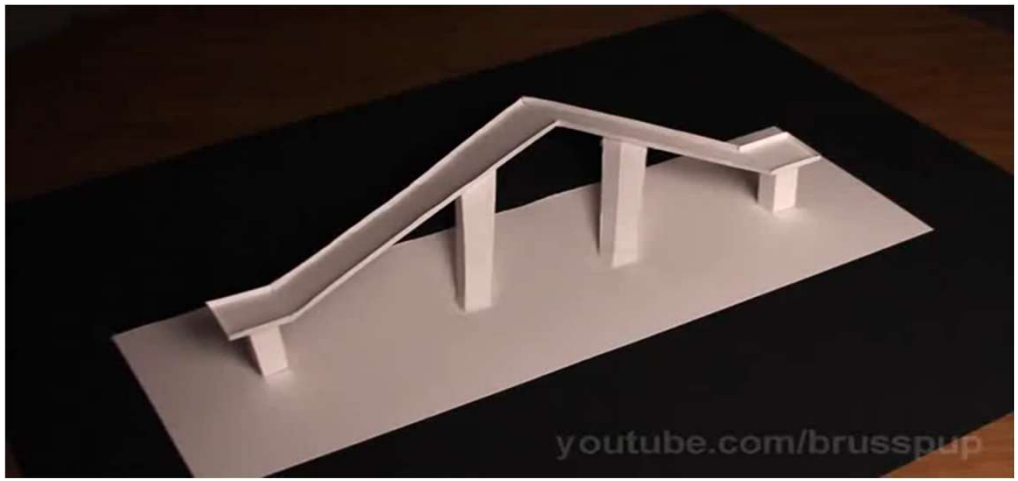
The main problem of vision:

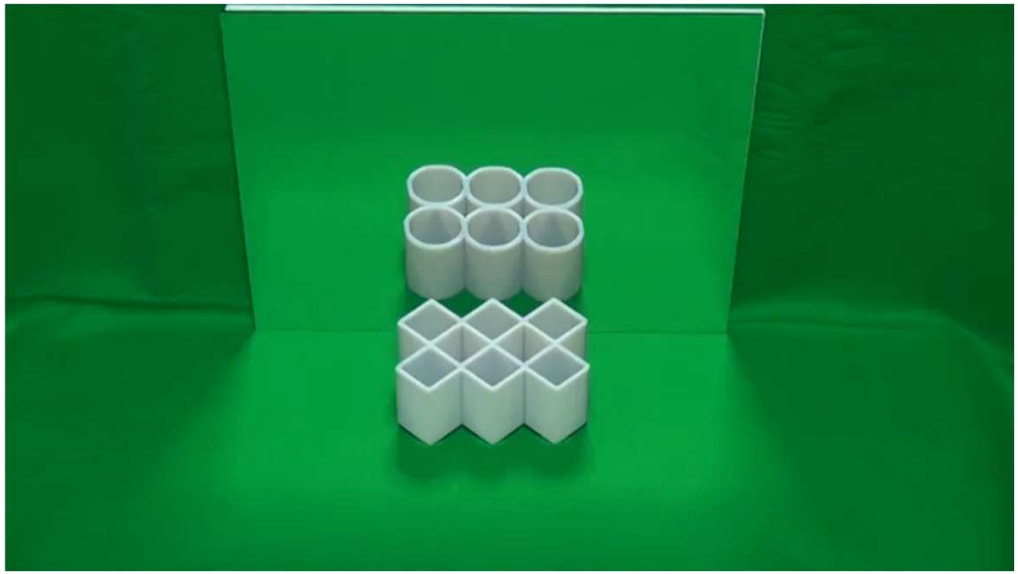
Even with a celebrated camera, recovery of structure is ill-defined



Geometry of Image Formation

The main problem of vision – recovery of structure is ill defined





Geometry of Image Formation

The main problem of vision – recovery of structure is ill defined